

Core Philosophy: “AI proposes, code verifies, and humans sign off where errors are irreversible or ambiguous.”

1. ESCALATION BOUNDARY MATRIX (MIRRORING BRIEF CRITERION 3)

Trigger / Scenario	Agent Action	Escalate?	Rationale & Boundary Defense
Column maps to 2 fields (ref)	Calls LLM for top-2 proposals (emp_id vs mgr_id)	Yes	Margin < 0.20; guessing destroys reporting hierarchy. Prompts consultant once.
Value cannot be cleaned ("L0A")	Proposes closest canonical status ("On Leave")	Yes	Unrecognized domain enum cannot be guessed; batched into 1 card, saved as rule.
Record fails validation twice	Validates → Auto-fixes → Re-validates failure	Yes	Missing required PII (e.g. email) cannot be synthesized; quarantined immediately.
Target Push 422 Client Error	Halts retry; parks record into Quarantine Ledger	Yes	Target rejects malformed payload; retries are wasteful. Requires schema review.
Sensitive conflict (Salary)	Withholds auto-merge; surfaces side-by-side diff	Yes	Discrepancy (e.g. ₹80,000 HRIS vs ₹88,000 Payroll); requires human sign-off.
Ambiguous dates (all days ≤ 12)	Holds format; prompts once per column (DD/MM vs MM/DD)	Yes	04/05/2021 is genuinely ambiguous. Blind guesses silently shift hire dates.
Near-duplicate identity	Fuzzy name match ≥ 90 + identical DOB	Yes	Merging two distinct people into one ID is irreversible data loss.
High-margin column mapping	Maps autonomously (conf ≥ 0.75, margin ≥ 0.20)	No	Statistically distinct match across aliases and types; verifiable against schema.
Cleanable validation format	Auto-fixes phone to E.164 (+91), casing to Title	No	Reversible; deterministic regex verification; zero information loss.
Target Push 503 Transient Error	Exponential backoff with jitter (max 3 retries)	No	Downstream network/service blip; succeeds silently on subsequent attempt.

2. DELTA SOLUTIONING & PRODUCTION ARCHITECTURE

• **Resolution Memory (Learned Rules Engine)**

- Human resolutions (mapping, enums, date format) are fingerprinted into the client-scoped rules table.
- On re-runs, learned rules fire *before* the escalation gate, auto-resolving repeat ambiguities.
- Review cards drop from 7 → 3 (-57%) on Run 2 on the planted fixture set.

• **Audit Trail, Idempotency & Rollback**

- Append-Only Event Spine:** SQLite triggers forbid UPDATE/DELETE on events; ensures deterministic SHA256 audit replay.
- Target Idempotency:** Idempotency-Key header prevents duplicate creates.
- Compensating Rollback:** Downstream outages trigger automated rollback (DELETE /stub/employees/{id}) to preserve target purity.

3. EMPIRICAL VERIFICATION & PRIVACY ARCHITECTURE

52 → 29	7 → 3 (-57%)	88.5% → 95.1%	22 / 22 Passed
SOURCE ROWS → MERGED EMPLOYEES	REVIEW CARDS (PLANTED FIXTURES)	AUTONOMY SCORE (RUN 1 → RUN 2)	AUTOMATED CI TESTS

*Autonomy Score Definition: (Automated Decisions / Total Pipeline Actions) × 100% measured across the 11 planted edge-case fixtures.

Enterprise Perimeter Isolation: The public hosted demo runs strictly on synthetic data with a cold-start “Reset Demo” control. In production enterprise deployments, local LLM inference (e.g. Ollama / on-prem vLLM) executes entirely inside the client VPC boundary with zero outbound data egress. Cross-tenant synonym matching is strictly opt-in and operates only on anonymized header tokens, never row-level PII.

4. PRODUCTION ROADMAP: WHAT I WOULD BUILD NEXT

- **Active Learning Feedback Loop:** Cluster unmapped column tokens across enterprise tenants to suggest industry-standard synonym maps while preserving strict tenant isolation (opt-in and headers-only).
- **Multi-Entity Relational Graph:** Link employees to Departments, Cost Centers, and Organizational Units with topological dependency ordering during the push stage.
- **Automated Schema Evolution:** Detect when client source exports drift (new columns or shifted formats) and generate automated diff migration proposals.